

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	18 June 2026
Team ID	
Project Name	ShopEZ – MERN Stack E-Commerce Platform
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Register a new account using email and password
FR-2	User Authentication	Login securely using registered credentials
FR-3	Product Listing	View all available products
FR-4	Product Search	Search products by product name
FR-5	Category Filtering	Filter products based on categories
FR-6	Wishlist Management	Add products to wishlist
FR-7	Shopping Cart	Add products to cart
FR-8	Shopping Cart	Update product quantity in cart
FR-9	Shopping Cart	Remove products from cart
FR-10	Order Management	Place orders from the shopping cart
FR-11	Order Management	View order history and order details
FR-12	Product Images	Display product images on product pages
FR-13	Admin Dashboard	Add new products
FR-14	Admin Dashboard	Edit and delete existing products
FR-15	Admin Dashboard	Manage customer orders and update order status

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	User-friendly interface with simple navigation and responsive design
NFR-2	Security	Secure authentication using JWT tokens and protected routes
NFR-3	Reliability	Consistent access to products, cart, wishlist, and order services
NFR-4	Performance	Fast loading of products, search results, and order information
NFR-5	Availability	Platform available to users with minimal downtime
NFR-6	Scalability	Ability to support increasing users, products, and orders
NFR-7	Maintainability	Modular MERN architecture for easier maintenance and updates
NFR-8	Compatibility	Supports modern browsers, desktops, tablets, and mobile devices
NFR-9	Data Integrity	Accurate storage and retrieval of user, product, cart, and order data
NFR-10	Responsiveness	Proper display and functionality across different screen sizes